# Data Science & Artificial Intelligence


## Artificial Intelligence

### Proposition and Predictive Logic

DPP 01 Discussion Notes


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[MCQ]

#Q. The number of different possible logical connectives on n variables is $2^{2^n}$ ______.

$\Downarrow$

logical fxn

A $2^n$

B $n^2$

C $2^{2n}$

D $2^{n2}$

[MCQ]

#Q. Consider the following statements:

S1: AI stands for Artificial Intelligence.

S2: $2 + 1 = 5$

• declarative and T/F

A. S1 is a proposition but S2 is not a proposition

B. S1 is not proposition but S2 is a proposition

C. Both S1 and S2 are propositions ✓

D. Neither S1 nor S2 is a proposition.

[MSQ]

#Q. The negation of the statement "We have those students who have applied for CS paper or DA paper" is:

$\sim[CS \vee DA] = \widetilde{CS} \wedge \widetilde{DA}$

A. We do not have those students who have applied for CS paper or DA paper

B. We have those students who have not applied for CS paper <u>and</u> not applied for DA paper

C. We have those students who have applied for CS paper and DA paper.

D. None of the above

[MCQ]

#Q. "If I am selected, then I will get the job." is equivalent to saying

$A \Rightarrow B \approx \sim B \Rightarrow \sim A$

A "If I am not selected, then I will not get the job".

B "If I will not get the job, then I am not selected."

C "If I will get the job then I am selected."

D None of the above

[MSQ]

#Q. The expression $P \leftrightarrow q$ can also be stated as.

A "p is necessary and sufficient for q"

B "if p then q, and conversely"

C "p exactly when q."

D $(p \rightarrow q) \wedge (q \rightarrow p)$

| $p\ q$ | $p \leftrightarrow q$ | $p \rightarrow q$ | $q \rightarrow p$ |
|---|---|---|---|
| 0 0 | 1 | 1 | 1 |
| 0 1 | 0 | 1 | 0 |
| 1 0 | 0 | 0 | 1 |
| 1 1 | 1 | 1 | 1 |

[MCQ]

#Q. Simplify the expression $(p \vee \neg q) \rightarrow (p \wedge q)$

**A** p

**B** q

**C** T

**D** F

| $p$ | $q$ | $(p \vee \neg q)$ | $p \wedge q$ | $(p \vee \neg q) \rightarrow (p \wedge q)$ |
|---|---|---|---|---|
| 0 | 0 | 1 | 0 | 0 |
| 0 | 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 |

[MCQ]

#Q. S1: $(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$

S2: $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p) \wedge (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$

A. S1 is satisfiable and S2 is not satisfiable

B. S2 is satisfiable and S1 is not satisfiable

C. Both S1 and S2 are satisfiable

D. Neither S1 nor S2 is satisfiable

Satisfiable ⇒ means at least one time T

| P Q R | P∨Q∨R | ¬P∨¬Q∨¬R |
|---|---|---|
| 0 0 0 | 0 | 1 |
| 0 0 1 | 1 | 1 |
| 0 1 0 | 1 | 1 |
| 0 1 1 | 1 | 1 |
| 1 0 0 | 1 | 1 |
| 1 0 1 | 1 | 1 |
| 1 1 0 | 1 | 1 |
| 1 1 1 | 1 | 0 |

S2: $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p) \wedge (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ = always F.

| zero@ | zero@ | zero@ | zero at | zero @ |
|---|---|---|---|---|
| PQR | PQR | PQR = 100 | PQR = 000 | PQR = 111 |
| 010 | 001 | = 110 | | |
| 011 | 101 | | | |

[MCQ]

#Q. The expression "$[\underbrace{(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)}_{}] \rightarrow r$" is :

G

zero @ PQR 000 001

zero @ PQR 100 110

zero @ PQR 010 011

A Tautology ✓

B Contradiction

C Contingency

D None of the above

| PQR | G | R | Result |
|---|---|---|---|
| 000 | 0 | 0 | 1 |
| 001 | 0 | 1 | 1 |
| 010 | 0 | 0 | 1 |
| 011 | 0 | 1 | 1 |
| 100 | 0 | 0 | 1 |
| 101 | 1 | 1 | 1 |
| 110 | 0 | 0 | 1 |
| 111 | 1 | 1 | 1 |
|  |  |  | 1 |

[MSQ]

#Q. Translate the statement: $\forall x(C(x) \vee \exists y(C(y) \wedge F(x, y)))$ into English, where $C(x)$ is "x studies Computer Science," $F(x, y)$ is "x and y are friends," and the domain for both x and y consists of all students in your school.

A. Every student in your school studies Computer Science or has a friend who studies Computer Science.

B. Every student in your school studies Computer Science or has at least one friend who studies Computer Science."

C. There exists a student who studies Computer Science, and all students are friends with this student.

D. Every student studies Computer Science and has at least one friend who studies Computer Science.

[NAT]

#Q. The number of expressions equivalent to $\forall w \neg \forall a \exists f (P(w, f) \wedge Q(f, a))$ are ____.

$\forall w \exists a \forall F (\neg P \vee \neg Q)$

$\forall w \exists a \forall F \neg(P \wedge Q)$

A: $\forall w \exists a \neg \exists f (P(w, f) \wedge Q(f, a))$

B: $\forall w \exists a \forall f \neg(P(w, f) \wedge Q(f, a))$

C: $\forall w \exists a \forall f (\neg P(w, f) \vee \neg Q(f, a))$

D: none

[MCQ]

#Q. The premises "It is raining today, and the roads are slippery.", "We will go hiking only if it is not raining", "If we do not go hiking, then we will visit the museum." and "If we visit the museum, then we will take the bus home." lead to the conclusion.

A We will not go hiking and we will take the bus home

B We will go hiking

C we will not take the bus home

D we do not visit the museum

A: it is Raining Today → T

B: Road are slippery → T

C: we will go hiking ×

d: we will visit museum → T

e: we take bus home → T

[MCQ]

#Q. The expression $(p \leftrightarrow q) \oplus (\neg p \leftrightarrow \neg r)$ is

**A** Tautology

**B** Contradiction

**C** Contingency

**D** None of the above

| P Q R | $P \leftrightarrow Q$ | $\neg P \leftrightarrow \neg R$ | EXOR |
|---|---|---|---|
| 0 0 0 | 1 | 1 | 0 |
| 0 0 1 | 1 | 0 | 1 |
| 0 1 0 | 0 | 1 | 1 |
| 0 1 1 | 0 | 0 | 0 |
| 1 0 0 | 0 | 0 | 0 |
| 1 0 1 | 0 | 1 | 1 |
| 1 1 0 | 1 | 0 | 1 |
| 1 1 1 | 1 | 1 | 0 |

[MCQ]

#Q. The expression $\neg(p \to q) \to \neg q$ is a.

**A** Tautology

**B** Contradiction

**C** Contingency

**D** None of the above

| P Q | $\neg(P \to Q)$ | $\neg Q$ | |
|---|---|---|---|
| 0 0 | 0 | 1 | 1 |
| 0 1 | 0 | 0 | 1 |
| 1 0 | 1 | 1 | 1 |
| 1 1 | 0 | 0 | 1 |

[MCQ]

#Q. Determine the truth value of each of the following statements where $U = \{1, 2, 3\}$ is the universal set :

(a) $\exists x \forall y, x^2 < y + 1$; (b) $\forall x \exists y, x^2 + y^2 < 12$ ;

for all x there is some y $x^2+y^2<12$

for some x , $\forall y$ $(x^2<y+1)$

A a is True; b is False

B a is False; b is True

C a is True; b is True ✓

D a is False; b is False

[NAT]

Zero @

#Q. Let $\alpha = (P \vee Q) \wedge (P \rightarrow Q) \wedge (Q \rightarrow P)$

0 0 0 1 0 1

The number of rows in the truth table having value of a as TRUE is $\underline{1}$.

| P | Q | |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

# THANK - YOU